



08-MAVZU · 42 TADAN · I BOB. HTML VA CSS

Box model va o'lcham birliklari. Flexbox va Grid

Har bir element — quti: content, padding, border, margin; box-sizing; px, %, em, rem, vw, vh; display; Flexbox bilan bir o'lchamli va CSS Grid bilan ikki o'lchamli sahifa maketlari.

3 soat · 1 nazariy + 2 amaliy

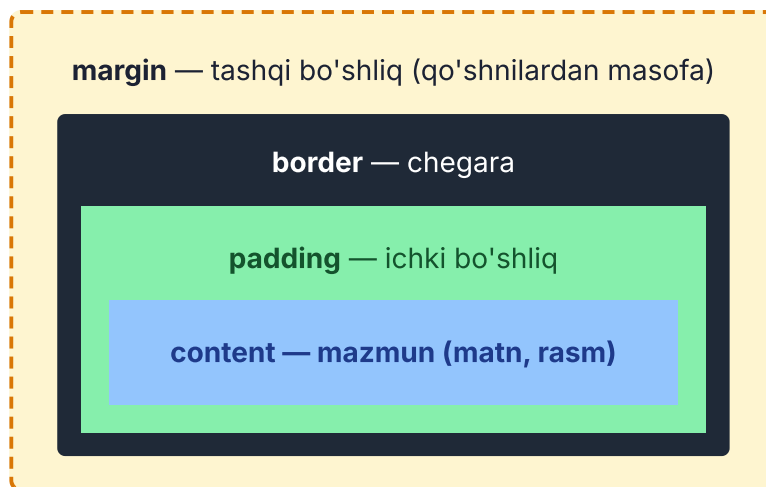
Bu darsda nimani o'rganasiz

- Brauzer har bir elementni qanday "quti" sifatida ko'rishini (box model);
- `margin`, `padding`, `border` va `box-sizing` ni;
- O'lcham birliklari: `px`, `%`, `em`, `rem`, `vw`, `vh`;
- **Flexbox** bilan menyu, kartalar qatori, markazlash;
- **CSS Grid** bilan butun sahifa maketi va galereya yaratishni.

Bu kursdagi eng muhim CSS mavzusi — 3 soat ajratilgan. Shoshilmang, har bir misolni ► tugmasi bilan o'zgartirib ko'ring.

Box model — har bir element quti

Brauzer sahifadagi **har bir** elementni (sarlavha, paragraf, rasm, tugma) to'rtburchak quti sifatida chizadi. Bu quti 4 qatlamdan iborat:



O'xshatish: rasmlı ramka

Devordagi rasmni tasavvur qiling. **Content** — rasmning o'zi. **Padding** — rasm atrofıdagi oq qog'oz hoshiya (paspartu). **Border** — yog'och ramka. **Margin** — devorda shu ramka bilan qo'shni ramka orasidagi bo'sh joy.

HTML

```
<style>
  .quti {
    width: 250px;
    padding: 20px;          /* ichki bo'shliq — fon rangi shu yerga ham
tarqaladi */
    border: 5px solid #4f46e5;
    margin: 30px;          /* tashqi bo'shliq — har doim shaffof */
    background: #e0e7ff;
  }
</style>
<div class="quti">Men qutiman. F12 bosib, meni tanlang va "Computed" panelidagi
box model chizmasiga qarang!</div>
<div class="quti">Men ikkinchi qutiman</div>
```

padding va margin yozish usullari

CSS

```
padding: 20px;                /* 4 tomon ham 20px */
padding: 10px 20px;           /* tepa-past 10px, chap-o'ng 20px */
padding: 10px 20px 30px;      /* tepa 10, chap-o'ng 20, past 30 */
padding: 10px 20px 30px 40px; /* tepa, o'ng, past, chap (soat streлки bo'yicha) */
padding-top: 10px;            /* faqat bitta tomon: -top, -right, -bottom, -left */

margin: 0 auto;               /* tepa-past 0, chap-o'ng auto → blokni GORIZONTAL MARKAZLASH */
```



Blokni markazga qo'yish

Kengligi berilgan blokni sahifa o'rtasiga qo'yish uchun: `width: 600px; margin: 0 auto; .`
Brauzer chap va o'ng bo'shliqni teng taqsimlaydi.

box-sizing — eng muhim bitta qator

Yuqoridagi `.quti` ning kengligi aslida necha piksel? `width: 250px` deb yozdik, lekin brauzer standart holatda **padding va border'ni qo'shib** hisoblaydi:

`250 + 20 + 20 (padding) + 5 + 5 (border) = 300px` 😊

Bu hisob-kitobni juda chalkashtiradi. Yechim:

CSS

```
/* Har bir loyihaning CSS faylini shu qoida bilan boshlang */
* {
  box-sizing: border-box; /* endi width = padding + border + content. 250px — haqiqatan 250px */
}
```

HTML

```
<style>
  div { width: 250px; padding: 20px; border: 5px solid; margin: 10px 0; }
  .content-box { box-sizing: content-box; border-color: tomato; }
  .border-box { box-sizing: border-box; border-color: green; }
</style>
<div class="content-box">content-box: men 300px kenglikdaman</div>
<div class="border-box">border-box: men aynan 250px kenglikdaman</div>
```

display xossasi

Qiymat	Xatti-harakat
block	Yangi qatordan, butun kenglik; width / height ishlaydi (div, p, h1)
inline	Qatorda, faqat mazmuncha; width / height va tepa-past margin ishlamaydi (span, a)
inline-block	Qatorda turadi, lekin width / height ishlaydi (tugmalar uchun qulay)
none	Element butunlay yashiriladi va joy egallamaydi
flex / grid	Zamonaviy maket tizimlari (pastda)

O'lcham birliklari

Birlik	Nimaga nisbatan	Misol	Qachon ishlatish
px	Mutlaq (ekran pikseli)	border: 1px	Chegaralar, soyalar, mayda detallar
%	Ota element o'lchamiga	width: 50%	Kengliklar

em	Joriy elementning shrift o'lchamiga	padding: 0.5em 1em	Matn o'lchamiga mos padding (tugmalar)
rem	Sahifa asosiy shriftiga (odatda 16px)	font-size: 1.5rem = 24px	Shriftlar va bo'shliqlar
vw	Ekran kengligining 1%	width: 100vw	Ekran kengligida bloklar
vh	Ekran balandligining 1%	min-height: 100vh	To'liq ekranli bosh qism (hero)

HTML

```
<style>
  html { font-size: 16px; }
  .ota { width: 400px; background: #e0e7ff; padding: 10px; }
  .yarim { width: 50%; background: #a5b4fc; } /* 400 ning 50% =
200px */
  .rem { font-size: 1.5rem; } /* 16 × 1.5 = 24px */
  .em-ota { font-size: 20px; }
  .em { font-size: 2em; } /* 20 × 2 = 40px */
  .ekran { height: 20vh; background: #fbcfe8; margin-top: 10px; }
</style>
<div class="ota"><div class="yarim">50% kenglik</div></div>
<p class="rem">1.5rem = 24px</p>
<div class="em-ota">Ota 20px – <span class="em">2em</span></div>
<div class="ekran">Ekran balandligining 20%i. Oynani kichraytirib ko'ring.
</div>
```

Flexbox — elementlarni bir qatorga terish

Flexbox — elementlarni **bir yo'nalishda** (qator yoki ustun) joylashtirish tizimi. Navigatsiya menyusi, tugmalar qatori, kartalarni yonma-yon qo'yish, biror narsani markazga joylash — Flexbox'ning ishi.

Ikki asosiy tushuncha

- **Flex konteyner** — `display: flex` berilgan **ota** element.
- **Flex elementlar** — uning **bevosita farzandlari**. Ular avtomatik ravishda bir qatorga tiziladi.

```
<style>
  .konteyner {
    display: flex;                /* ← shu bitta qator hammasini o'zgartiradi */
    gap: 10px;                   /* elementlar orasidagi masofa */
    background: #f1f5f9;
    padding: 10px;
  }
  .element { background: #4f46e5; color: white; padding: 20px; border-radius:
8px; }
</style>
<div class="konteyner">
  <div class="element">1</div>
  <div class="element">2</div>
  <div class="element">3</div>
</div>
<p>display: flex ni o'chirib ko'ring – elementlar yana ustma-ust tushadi.</p>
```

Asosiy va kesishgan o'q



O'xshatish: avtobus bekati

Odamlar bekatda navbatga turdi. Navbat yo'nalishi — **asosiy o'q** (main axis): `justify-content` odamlarni shu chiziq bo'ylab taqsimlaydi (boshida to'planish, o'rtada, oralari teng...). Navbatga perpendikulyar yo'nalish — **kesishgan o'q** (cross axis): `align-items` odamlarni bo'yi bo'yicha tekislaydi (tepasi, o'rtasi, pastki qismi bo'yicha).

Konteyner xossalari

HTML

```
<style>
.k { display: flex; background: #f1f5f9; margin: 8px 0; padding: 8px; height:
90px; border-radius: 8px; }
.k div { background: #4f46e5; color: #fff; padding: 8px 14px; border-radius:
6px; }
.k div:nth-child(2) { padding: 22px 14px; } /* ikkinchisi balandroq */

.k1 { justify-content: flex-start; } /* boshida (standart) */
.k2 { justify-content: center; } /* markazda */
.k3 { justify-content: flex-end; } /* oxirida */
.k4 { justify-content: space-between; } /* chetlarga yopishgan, oralari
teng */
.k5 { justify-content: space-around; } /* atrofida teng bo'shliq */
.k6 { justify-content: space-evenly; } /* barcha bo'shliqlar teng */
.k7 { align-items: center; } /* kesishgan o'q: vertikal markaz
*/
.k8 { align-items: flex-end; } /* pastga tekislash */
.k9 { justify-content: center; align-items: center; } /* ✨ to'liq
markazlash */
</style>
<div class="k k1"><div>flex-start</div><div>2</div><div>3</div></div>
<div class="k k2"><div>center</div><div>2</div><div>3</div></div>
<div class="k k3"><div>flex-end</div><div>2</div><div>3</div></div>
<div class="k k4"><div>space-between</div><div>2</div><div>3</div></div>
<div class="k k5"><div>space-around</div><div>2</div><div>3</div></div>
<div class="k k6"><div>space-evenly</div><div>2</div><div>3</div></div>
<div class="k k7"><div>align-items: center</div><div>2</div><div>3</div></div>
<div class="k k8"><div>align-items: flex-end</div><div>2</div><div>3</div>
</div>
<div class="k k9"><div>to'liq markaz</div><div>2</div><div>3</div></div>
```

Xossa (konteynerga)	Qiymatlar	Vazifasi
flex-direction	row column row-reverse column-reverse	Asosiy o'q yo'nalishi (qator yoki ustun)

<code>justify-content</code>	<code>flex-start</code> , <code>center</code> , <code>flex-end</code> , <code>space-between</code> , <code>space-around</code> , <code>space-evenly</code>	Asosiy o'q bo'yicha taqsimlash
<code>align-items</code>	<code>stretch</code> (standart), <code>flex-start</code> , <code>center</code> , <code>flex-end</code>	Kesishgan o'q bo'yicha tekislash
<code>flex-wrap</code>	<code>nowrap</code> <code>wrap</code>	Sig'masa keyingi qatorga o'tsinmi
<code>gap</code>	<code>16px</code>	Elementlar orasidagi masofa



`flex-direction: column` qilinsa, o'qlar **almashadi**: endi `justify-content` vertikal, `align-items` esa gorizontal ishlaydi.

Element xossalari: `flex`

HTML

```
<style>
.k { display: flex; gap: 8px; background: #f1f5f9; padding: 8px; margin: 8px 0; }
.k div { background: #4f46e5; color: #fff; padding: 14px; border-radius: 6px; }
.yon { flex: 0 0 150px; background: #f97316 !important; } /* qat'iy 150px */
.asosiy { flex: 1; } /* qolgan BARCHA joyni oladi */
.ikki { flex: 2; } /* flex:1 dan 2 barobar ko'p joy */
</style>
<div class="k"><div class="yon">Yon panel 150px</div><div class="asosiy">Asosiy qism – flex: 1</div></div>
<div class="k"><div class="asosiy">flex: 1</div><div class="ikki">flex: 2 (ikki barobar keng)</div><div class="asosiy">flex: 1</div></div>
```

CSS Grid — panjara (to'r) maketi

Grid — elementlarni **qator va ustunlardan** iborat to'rga joylashtiradi (ikki o'lchamli). Butun sahifa maketi, rasm galereyasi, mahsulotlar katalogi uchun ideal.



O'xshatish: shaxmat taxtasi yoki Excel jadvali

Grid — bu katakli taxta. Siz avval nechta ustun va qator bo'lishini belgilaysiz, keyin elementlarni kataklarga joylashtirasiz. Ba'zi elementlar bir nechta katakni egallashi mumkin — Excel'dagi kataklarni birlashtirish kabi.

HTML

```
<style>
.grid {
  display: grid;
  grid-template-columns: 1fr 1fr 1fr; /* 3 ta teng ustun */
  gap: 12px;
}
.grid div { background: #10b981; color: white; padding: 24px; border-radius:
8px; text-align: center; font-weight: bold; }
</style>
<div class="grid">
  <div>1</div><div>2</div><div>3</div>
  <div>4</div><div>5</div><div>6</div>
  <div>7</div>
</div>
```

fr birligi va repeat()

- `fr` (*fraction*) — **bo'sh joyning ulushi**. `1fr 2fr` — ikkinchi ustun birinchisidan 2 barobar keng.
- `repeat(4, 1fr)` = `1fr 1fr 1fr 1fr`.
- Aralash: `grid-template-columns: 250px 1fr;` — chapda qat'iy 250px yon panel, qolgani asosiy qism.

✨ Sehrli qator: avtomatik moslashuvchi galereya

HTML

```
<style>
  .galereya {
    display: grid;
    grid-template-columns: repeat(auto-fill, minmax(150px, 1fr));
    gap: 10px;
  }
  .galereya img { width: 100%; height: 120px; object-fit: cover; border-radius:
8px; }
</style>
<div class="galereya">
  
  
  
  
  
  
  
  
</div>
<p>"🎮 Yangi oynada" tugmasini bosib, oyna kengligini o'zgartiring – ustunlar
soni o'zi o'zgaradi!</p>
```

`repeat(auto-fill, minmax(150px, 1fr))` — "har bir ustun kamida 150px bo'lsin, nechta sig'sa shuncha ustun qil, ortgan joyi teng taqsimla". Telefonda 2 ta, planshetda 4 ta, kompyuterda 6 ta ustun — hech qanday qo'shimcha kodsiz.

Elementni bir nechta katakka cho'zish

HTML

```
<style>
.g { display: grid; grid-template-columns: repeat(4, 1fr); gap: 10px; }
.g div { background: #8b5cf6; color: #fff; padding: 20px; border-radius: 8px;
text-align: center; }
.katta { grid-column: span 2; grid-row: span 2; background: #f97316
!important; }
.keng { grid-column: 1 / 5; background: #0ea5e9 !important; } /* 1-
chiziqdan 5-chiziqqacha = 4 ustun */
</style>
<div class="g">
  <div class="katta">2x2 katta</div>
  <div>b</div><div>c</div><div>d</div><div>e</div>
  <div class="keng">Butun qator bo'ylab</div>
</div>
```

Nomlangan hududlar — sahifa maketi

HTML

```
<style>
  * { box-sizing: border-box; }
  body { margin: 0; font-family: sans-serif; }
  .sahifa {
    display: grid;
    grid-template-columns: 200px 1fr;
    grid-template-rows: 70px 1fr 50px;
    grid-template-areas:
      "header header"
      "aside main"
      "footer footer";
    min-height: 100vh;
    gap: 8px;
  }
  header { grid-area: header; background: #4f46e5; }
  aside { grid-area: aside; background: #a5b4fc; }
  main { grid-area: main; background: #e0e7ff; }
  footer { grid-area: footer; background: #312e81; }
  .sahifa > * { padding: 16px; color: #111; font-weight: bold; }
  header, footer { color: white !important; }
</style>
<div class="sahifa">
  <header>header</header>
  <aside>aside (yon panel)</aside>
  <main>main — asosiy mazmun</main>
  <footer>footer</footer>
</div>
```

`grid-template-areas` — maketni so'zlar bilan "chizish". Har bir qo'shtirnoq — bitta qator, har bir so'z — bitta ustun.

Flexbox yoki Grid?

	Flexbox	Grid
O'lcham	Bir o'lchamli: qator yoki ustun	Ikki o'lchamli: qator va ustun

Eng yaxshi joyi	Menyu, tugmalar, kartaning ichi, markazlash	Sahifa maketi, galereya, katalog
Qoida	Ular raqobatchi emas! Sahifa skeleti — Grid, uning ichidagi kichik qismlar — Flexbox.	

Amaliy mashg'ulot: portfolio sayti maketi (2-loyiha)

- 1 Yangi papka: `portfolio/`, ichida `index.html` va `style.css`.
- 2 HTML tuzilmasi (semantik!):

```

<body>
  <header class="header">
    <a class="logo" href="#">Ali<span>.dev</span></a>
    <nav class="menu">
      <a href="#haqida">Men haqimda</a>
      <a href="#loyihalar">Loyihalar</a>
      <a href="#aloqa">Aloqa</a>
    </nav>
  </header>

  <main>
    <section class="hero">
      <h1>Salom, men Ali – yosh veb-dasturchi</h1>
      <p>HTML, CSS va JavaScript o'rganyapman.</p>
      <a class="tugma" href="#loyihalar">Ishlarimni ko'rish</a>
    </section>

    <section id="loyihalar">
      <h2>Loyihalarim</h2>
      <div class="loyihalar">
        <article class="loyiha"><h3>Vizitka sayti</h3><p>HTML va CSS</p>
      </article>
        <article class="loyiha"><h3>Dars jadvali</h3><p>HTML
jadvallar</p></article>
        <article class="loyiha"><h3>O'quvchi kartasi</h3><p>CSS
dizayn</p></article>
        <article class="loyiha"><h3>Kalkulyator</h3><p>Tez kunda...</p>
      </article>
    </div>
  </section>
</main>

  <footer class="footer">© 2026 Ali</footer>
</body>

```

3

CSS — har bir qoidani o'qib, nima qilayotganini tushunib yozing:

```
<style>
* { box-sizing: border-box; margin: 0; padding: 0; }
body { font-family: "Segoe UI", Arial, sans-serif; color: #1f2937; line-height: 1.6; }

/* HEADER: logo chapda, menyu o'ngda — Flexbox */
.header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 16px 40px;
  background: white;
  box-shadow: 0 2px 10px rgba(0,0,0,.06);
}
.logo { font-size: 1.5rem; font-weight: 800; color: #111; text-decoration: none; }
.logo span { color: #4f46e5; }
.menu { display: flex; gap: 24px; }
.menu a { color: #374151; text-decoration: none; font-weight: 500; }

/* HERO: to'liq markazlash — Flexbox ustun bo'yicha */
.hero {
  min-height: 60vh;
  display: flex;
  flex-direction: column;
  justify-content: center;
  align-items: center;
  text-align: center;
  gap: 16px;
  padding: 40px 20px;
  background: linear-gradient(135deg, #eef2ff, #fdf2f8);
}
.hero h1 { font-size: 2.4rem; max-width: 700px; }
.tugma { background: #4f46e5; color: white; padding: 12px 26px; border-radius: 999px; text-decoration: none; font-weight: 600; }

/* LOYIHALAR: avtomatik moslashuvchi Grid */
#loyihalar { max-width: 1100px; margin: 0 auto; padding: 60px 20px; }
#loyihalar h2 { text-align: center; margin-bottom: 30px; font-size: 2rem; }
.loyihalar {
  display: grid;

```

```

    grid-template-columns: repeat(auto-fill, minmax(220px, 1fr));
    gap: 20px;
}
.loyiha {
    padding: 24px;
    border-radius: 14px;
    background: white;
    border: 1px solid #e5e7eb;
    box-shadow: 0 6px 20px rgba(0,0,0,.05);
}
.loyiha h3 { color: #4f46e5; margin-bottom: 6px; }

.footer { text-align: center; padding: 24px; background: #111827; color:
#9ca3af; }
</style>
<header class="header">
    <a class="logo" href="#">Ali<span>.dev</span></a>
    <nav class="menu"><a href="#haqida">Men haqimda</a><a
href="#loyihalar">Loyihalar</a><a href="#aloqa">Aloqa</a></nav>
</header>
<main>
    <section class="hero">
        <h1>Salom, men Ali – yosh veb-dasturchi</h1>
        <p>HTML, CSS va JavaScript o'rganyapman.</p>
        <a class="tugma" href="#loyihalar">Ishlarimni ko'rish</a>
    </section>
    <section id="loyihalar">
        <h2>Loyihalarim</h2>
        <div class="loyihalar">
            <article class="loyiha"><h3>Vizitka sayti</h3><p>HTML va CSS</p>
        </article>
            <article class="loyiha"><h3>Dars jadvali</h3><p>HTML jadvallar</p>
        </article>
            <article class="loyiha"><h3>O'quvchi kartasi</h3><p>CSS dizayn</p>
        </article>
            <article class="loyiha"><h3>Kalkulyator</h3><p>Tez kunda...</p>
        </article>
        </div>
    </section>
</main>
<footer class="footer">© 2026 Ali</footer>

```

- 4 "Men haqimda" bo'limini qo'shing: chapda rasmingiz, o'ngda matn — **Flexbox** bilan (`display: flex; gap: 30px; align-items: center`).
- 5 "Ko'nikmalarim" bo'limini qo'shing: HTML, CSS, JavaScript, Python... nishonlari (badge) — `flex-wrap: wrap` bilan.
- 6 F12 → Elements'da `display: grid` yonidagi **grid** belgisini bosing — brauzer panjarani chizib ko'rsatadi. Flex uchun ham xuddi shunday.

Ko'p uchraydigan xatolar

- ❌ `display: flex` ni **farzandga** yozish — u **otaga** (konteynerga) yoziladi.
- ❌ `justify-content: center` vertikal markazlamayapti — vertikal uchun `align-items` kerak (va konteynerning balandligi bo'lishi kerak!).
- ❌ `box-sizing: border-box` ni unutish — elementlar kutilganidan kengroq chiqib, qatorga sig'may qoladi.
- ❌ `inline` elementga (`a`, `span`) `width` / `height` berish — ishlamaydi; `display: inline-block` yoki `block` kerak.
- ❌ Grid'da `grid-template-columns: 1fr, 1fr` — vergul qo'yilmaydi: `1fr 1fr`.

O'zingizni tekshiring

? Element chegarasi bilan mazmuni orasidagi bo'shliq nima deyiladi?

- A. margin
- B. border
- C. padding
- D. gap

? `width: 200px; padding: 10px; border: 5px solid;` + `box-sizing: border-box`. Elementning to'liq kengligi?

- A. 230px
- B. 200px
- C. 215px
- D. 210px

? Menyuda logotipni chapga, havolalarni o'ngga surish uchun konteynerga nima yoziladi?

- A. `display: flex; justify-content: space-between`
- B. `display: flex; align-items: center`
- C. `display: grid; gap: 10px`
- D. `text-align: justify`

? Root shrift 16px bo'lsa, `2rem` necha piksel?

- A. 2px
- B. 16px
- C. 18px
- D. 32px

? `grid-template-columns: 1fr 3fr;` — konteyner 800px bo'lsa (gap yo'q), birinchi ustun necha piksel?

- A. 400px
- B. 100px
- C. 200px
- D. 267px

? Butun sahifa maketi (header, yon panel, asosiy qism, footer) uchun eng qulay vosita?

- A. Faqat Flexbox
- B. CSS Grid (`grid-template-areas`)
- C. HTML jadval
- D. `float`

Uyga vazifa

- Flexbox Froggy (24 daraja) va Grid Garden (28 daraja) o'yinlarini to'liq o'ting.
- Portfolio saytingizni yakunlang: "Men haqimda" (Flex), "Loyihalar" (Grid), "Ko'nikmalar" (flex-wrap), "Aloqa" (forma) va footer.
- Rasm galereyasi sahifasi yarating: 12 ta rasm, ulardan 2 tasi `span 2` bilan katta.

Yodda tuting

- Box model: content + padding + border + margin. Har doim `* { box-sizing: border-box; }`.
- Birliklar: `px` (aniq), `%` (otaga), `rem` (shriftlar), `vw/vh` (ekranga).
- Flexbox (konteynerga `display: flex`): `justify-content` — asosiy o'q, `align-items` — kesishgan o'q, `gap`, `flex: 1`.

- Grid: `grid-template-columns: repeat(auto-fill, minmax(200px, 1fr))`, `span`, `grid-template-areas`.

☐ Darsni o'qib bo'ldim

✓ Test javoblari

1. Element chegarasi bilan mazmuni orasidagi bo'shliq nima deyiladi? — **C) padding**
padding — ichki, margin — tashqi bo'shliq.
2. width: 200px; padding: 10px; border: 5px solid; + box-sizing: border-box. Elementning to'liq kengligi? — **B) 200px**
border-box'da width padding va border'ni o'z ichiga oladi.
3. Menyuda logotipni chapga, havolalarni o'ngga surish uchun konteynerga nima yoziladi? — **A)**
display:flex; justify-content: space-between
4. Root shrift 16px bo'lsa, 2rem necha piksel? — **D) 32px**
5. grid-template-columns: 1fr 3fr; — konteyner 800px bo'lsa (gap yo'q), birinchi ustun necha piksel? — **C) 200px**
Jami 4 ulush: $800/4 = 200\text{px}$; ikkinchisi 600px.
6. Butun sahifa maketi (header, yon panel, asosiy qism, footer) uchun eng qulay vosita? — **B) CSS Grid**
(grid-template-areas)